1 The nature of technology: Robots



You will learn

- what a robot is and how it works
- what control systems can and cannot do
- what robots can do now and how they are developing.

People have been fascinated by robots for hundreds of years. Fictional robots appear in science fiction films such as *Star Wars*. Robots appear in books and in video games. Today, robots are with us in real life. Robots are used to make electronics equipment such as TVs and computers. They are used to build cars. Robots drive cars. We live in an age of robots.



Talk about...

Some robots are designed to look like people. Do you think that makes them more likeable? Or do you prefer robots that don't look like people? Think of examples of robots in films and TV shows. Which ones do you like?

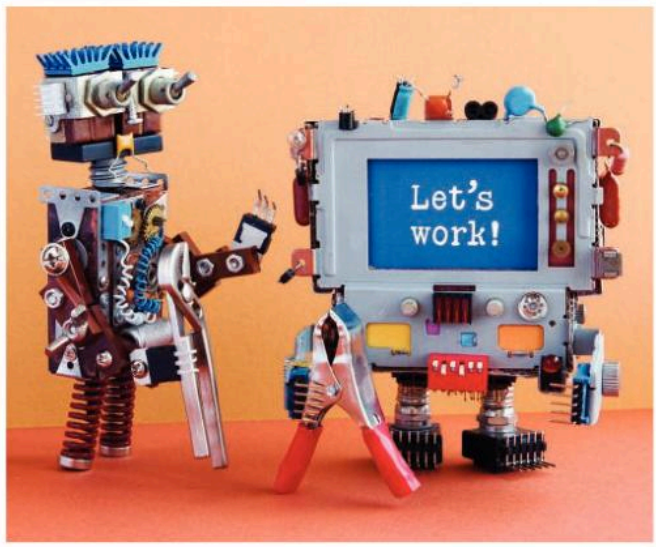


Learning outcomes: Describe the technology of robotics and control systems; Discuss the potential and the limitations of robotics and control systems

Class activity

Design your perfect robot assistant. What will your robot look like? What special tools and powers will it have? What jobs will your robot assistant do that will make your life easier?

robot robot arm drone
actuator sensor
controller control system
artificial intelligence
robot car nanobot



Did you know?

A British scientist called Kevin Warwick had a computer chip implanted into the nerves of his arm. In an experiment that took place in 2002, he controlled a robot hand over the internet. As he moved his hand in the USA, a robot arm in the UK made the same movements. In the future, people who have lost limbs might have mechanical replacements that work just as well as the real thing.



1.1 What is a robot?

In this lesson

You will learn:

- what a robot is and what it looks like.

Spiral back

In Student Book 4, you learned that microprocessors are built into machines and devices that we use every day. Cars, refrigerators and TVs have built-in microprocessors. A microprocessor adds more power to a device and makes it easier to use.

One machine is made more powerful and useful by microprocessors than any other. That machine is the robot. In this unit, you will learn how robots work and what they do.

What is a robot?

A **robot** is a machine. Robots are built by people to do a job. A robot must be programmed by a human to do its job. Once a robot is programmed, it does its job without help from humans.

Unless it breaks down, a robot does its job automatically. It does not need a human to control its actions. A robot does not get bored with its job. It does not make mistakes. The only time a human is needed is if the robot develops a fault.

What does a robot look like?

When people think of a robot, they often imagine them to look like people. Robots in science fiction films are often shown with arms, legs, a body and a head. In the real world, it is different.



Humanoid robots

Humanoid means like a human. Some robots are made to look like people. The robots take part in competitions where they carry out day-to-day tasks just like humans would. They are set challenges such as climbing a flight of stairs and carrying drinks without spilling them.

These robots are used by scientists to develop methods that might be used in the future.

Robot arms

Most robots are built to do jobs that humans do. Humans use their hands and arms to do most jobs. To do the same jobs, a robot has to be able to move like a human. This is the reason that many robots look like a human arm.

A **robot arm** has joints that bend and twist. A robot arm has a hand to grip things with.



Non-humanoid robots

Many robots do a job that humans do without looking human. An example is a robot vacuum cleaner. Another example is a **robot car**. A device does not have to look like a human to be called a robot.



Activity

Invent your own robot which would help with cleaning. Think of a name for your robot. List the main functions of your robot and the advantages of using it to do the cleaning.



Extra challenge

Design a poster to advertise the new robot cleaner.

Be creative

Create an image of the robot cleaner and include it in your poster.

Think again

Is your robot cleaner humanoid or not? Why did you make that choice?

1.2 What robots are used for

In this lesson

You will learn:

- what kinds of jobs robots can do
- how robots are used to build cars.

What jobs can robots do?

Repetitive tasks

Robots do jobs that are repetitive. This means jobs where an action needs to be done in the same way many times. An example is bolting wheels onto a car. Eight thousand wheels are put onto cars every day in a typical car factory. Humans find this kind of work boring. They can get tired and make mistakes or injure themselves. Robots never get bored or make mistakes.

Detailed work

Some jobs involve tasks to be carried out with great accuracy. A single mistake can mean a whole product is spoiled and must be thrown away. Making microprocessors involves a lot of detailed and accurate work. Robots are used to make microprocessors and other electronic equipment such as TVs.

Dangerous work

Robots are used in places where it is dangerous for humans to work. For example, robots are used on space missions. Robots called Sojourner and Pathfinder were sent to explore the surface of Mars. The picture gives you an idea of what Sojourner looked like.

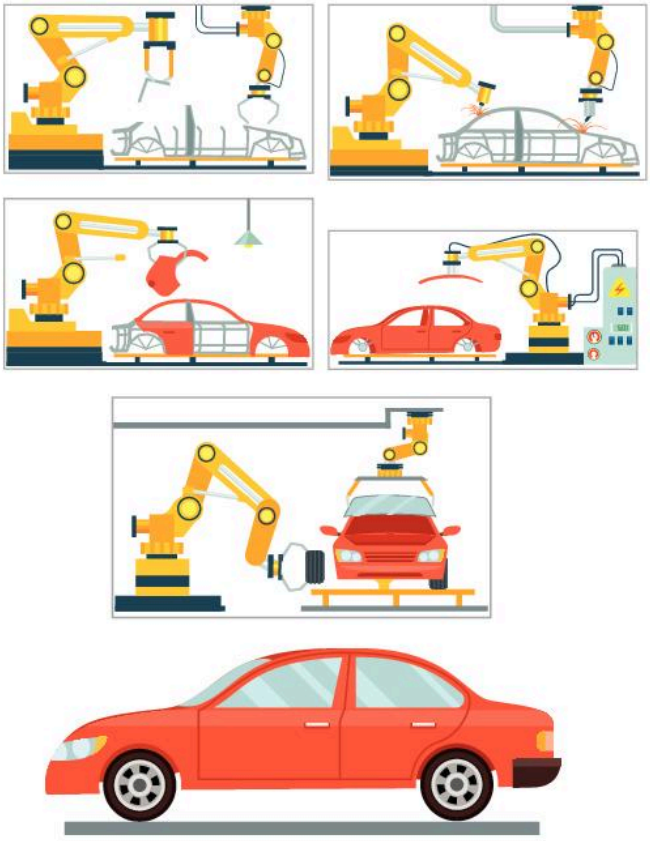


Robots are used under the sea. They explore the sea bed or inspect and repair oil rigs. Robots also work in environments where chemicals or radioactivity make it unsafe for humans to work.

Below are the main differences between how robots and humans do jobs.

Robots	Humans
• More accurate than humans	• Intelligent – they can find better ways of doing jobs
• Do not get bored or tired	• Creative – they can invent new things, such as robots
• Work faster than humans	• Solve problems
• Do not make mistakes	• Have emotions and feelings

Robots in the car industry



A car is not built by a single robot. The car moves along an assembly line. The car stops at points along the line. At each point, robots work together to build part of the car. For example, one team of robots will build the engine. Another team will put on the doors.

Robots must work together to make sure every task is performed in the right order. They must avoid colliding with each other. These teams are called **collaborative robots**. Collaborate means to work together.

Painting

Robots are used to paint cars. Robots always give cars a high-quality finish. The paints used on cars are hazardous to humans. Humans do not have to risk their health to do the job.



Activity

Make a poster with the title 'Robots versus humans'. List the strengths of robots and humans in your poster. Illustrate your poster.



Extra challenge

Describe a real-life example where robots are used to do dangerous jobs. Research the web to find your example. Some ideas for your research are fire-fighting, clearing landmines and inspecting pipelines. Are there images or videos of the robots in action?

Think again

Research how robots are used in the car industry. Find examples of jobs that robots are good at. Find examples of jobs that are still done by humans. Write up your findings in a report.



Digital citizen of the future

Some researchers think robots will replace 800 million jobs worldwide by 2030. Workers will need to train to do new jobs. Many of these jobs will be in the robotics industry. Having up-to-date computer skills will be more important than ever. Will you be ready for the new world of work?